

Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008

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Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008

UNION OF INDIA

India

Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008

Rule

PETROLEUM-AND-NATURAL-GAS-REGULATORY-BOARD-DETERMINATION-OF-NATURAL-GAS-PIPELINE-TARIFF-REGULATIONS-2008 of 2008

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Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008

Published vide Notification No. G.S.R. 807(E), dated 20th November, 2008

Last Updated 26th November, 2020

Petroleum and Natural Gas Regulatory Board

G.S.R. 807(E). - In exercise of the powers conferred by Section 61 of the Petroleum and Natural Gas Regulatory Board Act, 2006 (

19 of 2006

), the Petroleum and Natural Gas Regulatory Board hereby makes the following regulations, namely : -

1.

Short title and commencement.

(1)

These regulations may be called the Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008.

(2)

They shall come into force on the date of their publication in the Official Gazette.

2.

Definitions.

(1)

In these regulations, unless the context otherwise requires, -

(a)

"Act" means the Petroleum and Natural Gas Regulatory Board Act, 2006;

(b)

"appointed day" means the date of October 1, 2007;

(ba)

["approved tariff" means zone-wise natural gas pipeline tariff determined under regulation 4 or zone-wise tariff as per regulation 11 of the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008;]

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(c)

"Board" means the Petroleum and Natural Gas Regulatory Board established under sub-section (1) of section 3 of the Act;

(d)

"capacity of natural gas pipeline" means the capacity of natural gas pipeline as specified under the Petroleum and Natural Gas Regulatory Board (Authorizing Entities for Laying, Building, Operating or Expanding Natural Gas Pipelines) Regulations, 2008:

Provided that the capacity of natural gas pipeline in respect of entity authorized by the Central Government for laying, building, operating or expanding natural gas pipelines before the appointed day shall be as approved by the Board as per the basis specified in the relevant regulations for determining the capacity of natural gas pipeline;

(da)

["Imbalance Management Services" shall have the same meaning as assigned to it in the Petroleum and Natural Gas Regulatory Board (Imbalance Management Services) Regulations, 2016;]

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(e)

"initial unit natural gas pipeline tariff" means the unit natural gas pipeline tariff determined for a period commencing from -

(i)

the date of commissioning of the natural gas pipeline in any financial year and ending on the last day of that financial year in case the natural gas pipeline is commissioned on or after the notification of these regulations; or

(ii)

the date of coming of the natural gas pipeline within the purview of these regulations in any financial year and ending on the last day of that financial year in case the natural gas pipeline is commissioned before the notification of these regulations:

Provided that in case the period between the date of commissioning of the natural gas pipeline or the date of coming of the natural gas pipeline within the purview of these regulations in any financial year, as the case may be, and the last day of that financial year is less than one hundred and eighty three days, the initial unit natural gas pipeline tariff shall be for a period commencing from the date of commissioning of the natural gas pipeline or the date of coming of the natural

gas pipeline within the purview of these regulations in the financial year and extend up to the last day of the immediately succeeding financial year:

Provided further that the initial unit natural gas pipeline tariff shall be fixed on a provisional basis first and then finalized as per the basis and in the time frame specified under clause 9 of Schedule A.

(ea)

["invoicing entity" means the authorised entity laying, building, operating or expanding natural gas pipeline, where natural gas is evacuated by the shipper at unified exit point, or which is part of national gas grid system;

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(eb)

"MMBTU" means million British Thermal Units on gross calorific value ("GCV") basis;

(ec)

"national gas grid system" means network of all such natural gas pipelines within India which are fully interconnected with each other (including those which are partly commissioned and so interconnected) and as enlisted in Schedule C to these regulations and the Board may, by order, amend the said Schedule C from time to time;]

(f)

"natural gas pipeline" means any pipeline including spur lines for transport of natural gas and includes all connected equipments and facilities, such as, compressors, storage facilities, metering units but excludes -

(i)

dedicated pipeline laid to transport natural gas to a specific customer to meet his requirement and not for resale;

(ii)

pipelines in a city or local natural gas distribution network which are regulated by the Petroleum and Natural Gas Regulatory Board (Authoring Entities to Lay, Build, Operate or Expand City or Local Natural Gas Distribution Networks) Regulations, 2008;

(g)

"natural gas pipeline tariff" means the unit rate of tariff for a natural gas pipeline (excluding statutory taxes and levies) in rupees per million British Thermal Units (Rs./ MMBTU) for transport of natural gas;

(h)

["tariff review" means the review of the unit natural gas pipeline tariff after every five consecutive years by the Board with the first tariff review to be done after the end of five consecutive years after the end of the initial unit natural gas pipeline tariff period as specified under clause (e):

[Substituted by Notification F.No. PNGRB/COM/2-NGPL Tariff (3)/2019, dated 27.5.2019 (w.e.f. 20.11.2008).]

Provided that the unit natural gas pipeline tariff so determined at the time of any tariff review shall apply for the period up to the next tariff review:

Provided further that the gap between two tariff reviews shall not be less than two consecutive financial years after the end of the financial year in which last tariff fixation occurred:

Provided also that the tariff review can be done earlier in terms of provisions of sub-clause (9) of clause 9 of Schedule A.]

(i)

["tariff zone" means the zone-

[Substituted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(i)

of a length of three hundred kilometers each along the route of the natural gas pipeline from the point of injection to the end point:

Provided that the last zone of the natural gas pipeline may be of a length of three hundred kilometers or less:

Provided further that any natural gas pipeline of a length less than three hundred kilometers shall be counted as a zone; and

(ii)

a corridor along the natural gas pipeline with a width of up to ten percent of the total length of the natural gas pipeline without including the length of the spur lines or fifty kilometers measured from the nearest point on the surface of the

natural gas pipeline on both sides, and includes the point of origin and the end point of the natural gas pipelines, whichever is less, and -

(a)

the first tariff zone shall be counted with reference to any zone in which the point of injection of natural gas into the natural gas pipeline falls; and

(b)

the subsequent tariff zone or tariff zones, as the case may be, shall be counted separately on either side along the contractual path for delivery of natural gas in the natural gas pipeline;

Provided that natural gas pipeline tariff for transport of natural gas from the same source shall be uniform for all the customers located within the zone:

Provided further that the entity shall supply natural gas to any customer located in the zone subject to the techno-commercial feasibility of laying, building, operating or expanding a new spur line from the natural gas pipeline.

Explanation. - For the purposes of this clause, the entity shall submit to the Board information regarding various tariff zones and their sequential numbering in the natural gas pipeline from each point of injection.]

(j)

["tariff zones for unified tariff" means the -

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(i)

first tariff zone for unified tariff of a length of three hundred kilometres from the unified entry point on either side of the national gas grid system; or

(ii)

second tariff zone for unified tariff, being the remaining length of the national gas grid system on either side of first tariff zone for unified tariff:

(k)

"unified tariff" means unit natural gas pipeline tariff, in respect of the national gas grid system determined in accordance with regulation 5A, for transportation of natural gas (excluding statutory taxes and levies) denominated in rupees per MMBTU;

(l)

"unified contractual path" means the route on the national gas grid system between the unified entry point and the unified exit point for a shipper;

(m)

"unified entry point" means the point on the national gas grid system where a shipper injects natural gas;

(n)

"unified exit point" means the point on the national gas grid system from where a shipper evacuates natural gas.]

(2)

Words and expressions used and not defined in these regulations, but defined in the Act or in the rules or regulations made thereunder, shall have the meanings respectively assigned to them in the Act or in the rules or regulations, as the case may be.

3.

Application.

- These regulations shall apply to an entity -

(1)

authorized by the Central Government under regulation 17 of the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008 for laying, building, operating or expanding a natural gas pipeline before the appointed day;

(2)

laying, building, operating or expanding a natural gas pipeline before the appointed day and authorized by the Board for such activities under regulation 18 of the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008; and

(3)

laying, building, operating or expanding a dedicated pipeline which is converted into a natural gas pipeline under the provisions of sub-regulation (1) or (2) of regulation 19 of Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008.

(4)

[authorized by the Board pursuant to a policy directive issued by the Central Government under sub-section (2) of section 42 of the Act;

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(5)

authorised by the Board under regulation 9 of the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008 and whose tariff is fixed under regulation 11 of the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008, for the limited applicability of unified tariff in accordance with regulation 5A.]

4.

Determination of natural gas pipeline tariff.

- The natural gas pipeline tariff in respect of an entity covered under clause (1), clause (2) or clause (3) of regulation 3 shall be determined as per the procedure at Schedule A.

5.

Form for data submission by entity for determination of natural gas pipeline tariff.

- Every entity shall submit to the Board the financial costs and other data in the form at Attachment 1 of Schedule A, if, as on the day of the notification of these regulations, the natural gas pipeline -

(1)

is not in operation, then at least six months before the likely date of commissioning of the natural gas pipeline, or within ninety days of the date of notification of these regulations, whichever is later; or

(2)

is already in operation, then, in relation to an entity referred to in -

(a)

clause (1) of regulation 3, within ninety days of the notification of these regulations;

(b)

clause (2) of regulation 3, within ninety days of the grant of authorization by the Board; or

(c)

clause (3) of regulation 3, within ninety days of the date of authorization by the Board:

Provided that the entity shall charge initial unit natural gas pipeline tariff "on account basis" based on these regulations from the date of commissioning of natural gas pipeline referred to in clause (1) of regulation 3 or the date of authorization by the Board of natural gas pipeline referred to in clause (2) or clause (3) of regulation 3, as the case may be, till the date the Board provisionally fixes the initial unit natural gas pipeline tariff:

Provided further that the entity shall carry out adjustments, with a retrospective effect with the customers for the difference between the initial unit natural gas pipeline tariff that the entity had so charged and that provisionally fixed by the Board.

5A.

[Procedure for Determination of Unified Tariff and matters incidental thereto.

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(1)

Notwithstanding anything contained in these regulations, Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008 and any other regulation made under the Act and for the time being in force, invoicing entity shall raise invoice towards and collect the unified tariff for unified contractual path instead of the approved tariff from the shipper and the shipper shall be liable to pay such unified tariff to such invoicing entity.

(2)

The invoice referred to in sub-regulation (1), shall be raised for the quantity of gas delivered to the shipper by the

transporter at the unified exit point and the ship or for pay quantity: Provided that the entity laying, building, operating or expanding natural gas pipeline other than the one which is part of national gas grid system shall continue to charge the approved tariff from the shipper and the shipper shall continue to pay the approved tariff to such entity.

(3)

Revenue entitlement of the entities laying, building, operating or expanding natural gas pipelines which are part of the national gas grid system under unified tariff system, shall be the revenue with reference to approved tariff for such natural gas pipelines, that is to say that the entities shall remain revenue neutral and the procedure for determination of unified tariff shall be as specified in Part A of Schedule B of these regulations and the procedure for settlement amongst the pipeline entities shall be as specified in Part B of Schedule B.

(4)

Considering the experience and suggestions, if any, received from the stakeholders during the initial period of six months from the date of the commencement of the Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Second Amendment Regulation, 2020, the Board may, by order notified in the official gazette, suitably modify the provisions of this regulation, wherever considered necessary, and on the issue of such order, this regulation shall be deemed to be modified accordingly.]

6.

Miscellaneous.

- If any dispute arises with regards to the interpretation of any of the provisions of these regulations, the decision of the Board shall be final.

A

[see regulations 4 and 5]

Procedure for determination of natural gas pipeline tariff.

The natural gas pipeline tariff shall be determined by considering a reasonable rate of return on normative level of capital employed plus a normative level of operating expenses in the natural gas pipeline.

Explanation. - The expression "normative level" shall be with reference to a level, which is both reasonable and justifiable in terms of incurrence of capital and operating expenditure required for laying, building, operating or expanding an efficient natural gas pipeline over its economic life.

1. Financial feasibility. - The entity to which these regulations apply shall submit all technical, operating, financial and cost data of the natural gas pipeline project that may be required by the Board in determination of the natural gas pipeline tariff.

2. Methodology for determination of natural gas pipeline tariff. - The unit rate of natural gas pipeline tariff to be charged for a period shall be the calculated based on the "Discounted Cash Flow" (DCF) methodology¹ considering the reasonable rate of return as specified in clause 3 to be the project's internal rate of return. The parameters relevant to the applicability of the DCF methodology have been described in detail in clauses 4 to 6 below.

3. Reasonable rate of return. - The rate of return on capital employed shall be the rate of return on capital employed equal to twelve percent post-tax. The rate of return on capital employed once applied to a natural gas pipeline project shall remain fixed for the entire economic life of the project.

Note: - The pre-tax rate of return on capital employed shall be computed by grossing-up twelve percent by the nominal rate of income tax applicable for corporate assesses as per the provisions of the Income Tax Act, 1961, as amended from time to time.

[Explanation.

[Inserted by Notification No. F.No. PNGRB/COM/2-NGPL/Tariff (3)/2019 Vol-II, dated 27.3.2020 (w.e.f. 20.11.2008).]

- For removal of doubt, it is clarified that in case more than one nominal rates of income tax are available as an option to the entity under the law relating to income tax, the lowest of such nominal rates of income tax shall be considered for the purpose of grossing-up the post-tax rate of return of twelve per cent for computing pre-tax rate of return on capital employed.]

4. Return on total capital employed. - (1) The reasonable rate of return shall be applied on the total capital employed to determine the return on capital employed in the project over its economic life and the authorized entity is free to leverage the financing of the project in any suitable manner.

(2)

The total capital employed shall be equal to the gross fixed assets in the project less accumulated depreciation² plus

normative working capital (equal to thirty days of operating costs excluding depreciation and eighteen days natural gas pipeline tariff receivables).

(3)

The gross fixed assets shall be equal to their actual historical cost of acquisition (including the cost of any subsequent replacement or improvement or modification) or that normatively assessed by the Board, whichever is lesser and required in the natural gas pipeline project over its economic life based on the principles³ to create and sustain an efficient infrastructure namely:-

(a)

treatment of an investment in the fixed asset in determination of total capital employed shall be as per the basis indicated in Attachment 2;

(b)

capital costs in similar projects, if any, elsewhere in India benchmarked on a "like-to-like" basis;

(c)

appropriateness of the pipeline design and the operating philosophy with regards to maximum allowable operating pressure;

(d)

optimization of the equipments and facilities such as, compressors, metering systems, SCADA, fire fighting required, based on an assessment of the appropriate available technology;

(e)

spur lines;

(f)

design parameters for compressors; and

(g)

assessment of the costs of major equipments and facilities in the natural gas pipeline, laying or building costs, project management consultancy and pre-operative expenditure.

5. Operating costs. - Operating costs⁴ required in the operation and maintenance of the natural gas pipeline over its economic life shall be computed, on an actual basis or based on a normative assessment by the Board, whichever is lower, over the following functional cost heads, namely: -

(1)

consumables;

(2)

utilities -

(a)

power;

(b)

fuel (including the cost of natural gas and the natural gas pipeline tariff not recovered on the volume of system-use natural gas consumed in the natural gas pipeline);

(c)

water;

(3)

salaries and wages;

(4)

repairs and maintenance;

(5)

insurance premia on assets (excluding the value of loss of profit) and on line-pack volumes;

(6)

administrative overheads [to the extent not classifiable under sub-clause (1) to sub-clause (5)], related and also commensurate to the level of operations in the natural gas pipeline;

(7)

depreciation on fixed assets on straight line basis based on rates as per Schedule VI to the Companies Act, 1956); and

(8)

[Miscellaneous income realizable from a fixed asset included in the capital employed or out of an expense considered as an operating cost, including income from imbalance management services in accordance with the provisions of Petroleum and Natural Gas Regulatory Board (Imbalance Management Services) Regulations, 2016, but excluding interest income, profit or loss on sale or transfer of any fixed or other asset, shall be netted from the operating cost.]

[Substituted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

6. Volumes to be considered in determination of the unit natural gas pipeline tariff. - (1) The volumes of natural gas to be considered as divisor in the determination of the unit natural gas pipeline tariff over the economic life of the project shall be computed on a normative basis as indicated below:-

(a)

the divisor for each of the first five years of operations of the natural gas pipeline shall be arrived by multiplying the applicable percentage utilization for the year, as per the basis indicated below, with the sum of the capacity requirement of the entity and the firmed-up contracted capacity with other entities as specified under the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008:-

Year of natural gas pipeline operations

Percentage utilization

First

60%

Second

70%

Third

80%

Fourth

90%

Fifth

100%

(b)

the divisor for the sixth and the subsequent years of operation of the natural gas pipeline shall be equal to:

(i)

one hundred percent of the sum of the capacity requirements of the entity and the firmed-up contracted capacity with other entities as specified under the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008; and

(ii)

the actual volume of natural gas transported on common carrier basis against the extra capacity as specified under the Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008 during the immediately preceding year of operation of the natural gas pipeline.

Explanation. - The adjustment for the difference between the actual volume transported on common carrier basis in any year and that included in the divisor under sub-item (ii) to item (b) of sub-clause (1) shall be carried out on a prospective basis in the DCF calculations for unit natural gas pipeline tariff.

(c)

and (d) [-]

(e)

[notwithstanding anything contained in any other regulation including regulation 12 and regulation 21 of Petroleum and Natural Gas Regulatory Board (Authorizing Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008, in case it is proposed to expand the capacity of the natural gas pipeline by more than ten percent of that authorized by the Board, the entity shall submit a proposal for consideration of the Board and the Board may allow for expansion of the capacity in the natural gas pipeline, but the tariff of the pipeline so determined based on the capex, opex and volumes of expanded natural gas pipeline shall be capped at the levels of the applicable tariff based on the capex, opex and volumes of the pipeline without expansion.

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

It is clarified that any interconnection between two natural gas pipelines shall not be considered as extension or expansion or tie in connectivity.]

(2)

The volume of natural gas determined under sub-clause (1) shall be converted into its energy equivalence in MMBTU terms for the purpose of-

(a)

determination of the provisional unit natural gas pipeline tariff by considering the weighted average of heat value of different natural gases which are likely to be transported by the entity during this period;

(b)

determination of the final initial unit natural gas pipeline tariff by considering the weighted average heat value of the natural gas delivered to customers during the initial unit natural gas pipeline tariff period;

(c)

first tariff review by considering the weighted average heat value of natural gas delivered to the customers between the period from the end of the initial unit natural gas pipeline tariff period and the due date of the first tariff review;

(d)

any subsequent tariff review by considering the weighted average heat value of the natural gas delivered to the customers during the period between that tariff review and the immediate previous tariff review.

(3)

[Notwithstanding anything contained in any regulations made under the Act, 350 working days shall be considered in a year for the purpose of tariff determination under these regulations.]

[Substituted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

7. [Economic life.

[Substituted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

- Notwithstanding anything contained in any regulations made under the Act, economic life of the pipeline shall be considered as thirty years from the date of commissioning for the purpose of tariff determination under these regulations. In case, the natural gas pipeline has been authorized for more than thirty years or its authorisation has been subsequently extended beyond thirty years or is in operation beyond thirty years, the DCF model for the tariff computation may be made for the entire period of authorization or operation till the next tariff review, including the extended period. Any replacement of the pipeline, in excess of five per cent of the length of the pipeline including spur-lines and tie-in lines, before the completion of the aforesaid economic life, shall not be considered for tariff determination if approval of the Board has not been obtained by the entity for such replacement.]

8. Procedure for apportioning of unit natural gas pipeline tariff over the tariff zones. - [(1) The entity shall submit for the Board's approval, the calculations in respect of apportioning of the unit natural gas pipeline tariff over all the tariff zones during the economic life of the project in the form specified in Part II of Attachment 1(i), where X% shall be a single number lower than one hundred per cent and Y% shall be a single number lower than fifty per cent.

In case the entity fails to submit the required information within the time determined by the Board for such purpose, the Board may apportion the unit natural gas pipeline tariff over all the tariff zones in the ratio of tariff for various tariff zones during the immediate preceding zonal tariff determination in respect of that pipeline or in any other manner as may be deemed fit by the Board.]

[Substituted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

(2)

No adjustment shall be made by the entity with the customers for any over-achievement or under-achievement in the recovery of the natural gas pipeline tariff by the entity due to the volumes actually transported in different tariff zones being different than the volumes considered by the entity for apportioning the unit natural gas pipeline tariff for each of the tariff zones during-

(a)

the period of initial unit natural gas pipeline tariff;

(b)

the first five and consecutive years after the end of the initial unit natural gas pipeline tariff; and

(c)

the period between any two consecutive tariff reviews.

9.

Determination of initial unit natural gas pipeline tariff and review of unit natural gas pipeline tariff. - (1) The unit natural gas pipeline tariff shall be determined for the natural gas pipeline over its economic life and levelized during-

(a)

the period applicable for the initial unit natural gas pipeline tariff;

(b)

the period of five consecutive years after the end of the initial unit natural gas pipeline tariff; and

(c)

the period between any two consecutive tariff reviews.

(2)

The initial unit natural gas pipeline tariff including its apportioning over all tariff zones shall be determined based on provisional computations.

(3)

The final computations in respect of the initial unit natural gas pipeline tariff shall be made considering the adjustments required for-

(a)

the actual capital and operating costs or that normatively assessed by the Board, whichever is lower as specified under clause 4 and clause 5;

(b)

volume consideration as per item (a) of sub-clause (1) of clause 6; and

(c)

volume conversion into its energy equivalence as per item (b) of sub-clause (2) of clause 6.

(4)

[The adjustment on account of variation in the provisional initial unit natural gas pipeline tariff, final initial unit natural gas pipeline tariff and tariff determined under tariff review shall be made in the DCF calculations and the derived tariff shall be charged from the customers on prospective basis till next review, that is, the tariff shall be applicable from the first day of the month, following the month, in which the tariff order (zonal tariff in case of applicability of zonal tariff) is issued by the Board.]

[Substituted by Notification F.No. PNGRB/COM/2-NGPL Tariff (3)/2019, dated 27.5.2019 (w.e.f. 20.11.2008).]

(5)

The final unit natural gas pipeline tariff as specified under sub-clause (3) shall apply for the period upto the first tariff review until the time the same is revised by the Board;

(6)

The entity shall submit for Board's approval the data for determination of the unit natural gas pipeline tariff after the notification of these regulations in the form and time frame specified under regulation 5 alongwith the provisional computations of the initial unit natural gas pipeline tariff including the apportioning of the same over all tariff zones;

Provided that the final computations of the initial unit natural gas pipeline tariff including apportioning of the same over all tariff zones shall be submitted three months before the last date of the period applicable for the initial unit natural gas pipeline tariff.

(7)

The entity shall submit computation of the unit natural gas pipeline tariff including apportioning of the same over all tariff zones in the form specified under regulation 5 at the time of each tariff review within three months of the-

(a)

end of the last year of the five consecutive years after the initial unit natural gas pipeline tariff; or-

(b)

end of the preceding tariff review period:

Provided that the computation shall consider the actual cost, financial and operational data or that normatively assessed by the Board, whichever is lesser as per clauses 4 and 5 in respect of the period specified under sub-clause (a) or (b),

as the case may be.

(8)

Adjustments on account of the following variations during the five consecutive years after the initial unit natural gas pipeline tariff or any tariff review shall be made in the DCF calculations on a prospective basis by considering the remaining period of the economic life of the project, namely:-

(a)

the variations on account of the actual performance with respect to the capital and operating costs which when compared with the normative parameters identified under clauses 4 and 5;

(b)

the variations in respect of the volumes under Explanation to sub-item (ii) of sub-clause (1) of clause 6; and

(c)

the variations in the energy equivalence under sub-clause (2) to clause 6.

(9)

The Board may, either on its own or on the entity's request, carry out a review of the unit natural gas pipeline tariff any time between two tariff reviews, as the case may be, considering-

(i)

applicable nominal rate of income tax used for grossing-up the rate of return on capital employed;

(ii)

sudden change in any parameter used in the determination of the unit natural gas pipeline tariff.

(10)

The provisional and final initial unit natural gas pipeline tariff, unit natural gas pipeline tariff for the period of five consecutive financial years after the initial unit natural gas pipeline tariff and the unit natural gas pipeline tariff fixed after any tariff review thereafter shall continue to apply till the time the same is revised by the Board.

1. Discounted Cash Flow methodology refers to equating the inflows from the projected revenue earnings out of natural gas pipeline tariff with the outflows of capital and operating expenditures over the economic life of the project by discounting these flows at the project's reasonable rate of return. The volumes and outflows are estimated over the economic life which results in the determination of the natural gas pipeline tariff required to be earned by the project to achieve the internal rate of return.

2. Depreciation shall be calculated by applying the rates of depreciation as specified in Schedule VI to the Companies Act, 1956 on the value of gross fixed assets on a straight line basis.

3. The basis and parameters identified here are not exhaustive and may vary depending upon the specific natural gas pipeline project under consideration which the Board may apply in a transparent manner.

4. Operating costs are those which are necessary for a proper upkeep and maintenance of a tangible asset created for operation of the natural gas pipeline and does not include any financial cost, like, interest on loans, bad-debts, sales promotion, advertisement expenses (except for tenders), expenditure incurred in raising or servicing of capital or debenture or bond or any debt, exchange variation on revenue account, but includes normal bank charges, bank guarantee charges, LC charges and bank charges for bid and performance bonds required as per the terms of authorization.

Attachment 1

[see regulation 5]

Project capital cost summary (Capex outflows).

INR

FE

Sr. No.

Description

Indian rupees

Foreign exchange

Total (*)

A

Plant and machinery

1

Receiving terminal [see attachment 1 (a)]

2

Compressor [see attachment 1 (b)]

3

Intermediate terminal [see attachment 1 (c)]

4

Delivery terminal [see attachment 1 (d)]

5

Pipeline [see attachment 1 (e)]

Subtotal (A)

B

Engineering costs

1

Detailed engineering, procurement, construction, supervision
and project management

2

Service Tax on above

Subtotal (B)

C

Site related costs

1

Land

2

Any other (details to be given)

Subtotal (C)

D.

Owner's cost start-up expenses

1

Start-up and commissioning expenses

2

Owner's management expenses

Subtotal (D)

E

Other Costs

1

Contingencies and escalation

2

Preliminary and pre-operative expenses

3

Interest during construction period and financing charges

4

Margin money for working capital

Subtotal (E)

Total (A+B+C+D+E)

(*) INR + FE converted to INR

Note : Detailed assumption / calculations to be
given for them E.

Attachment 1(a)

[see regulation 5]

Capital cost of receiving terminal

Sr. No.

Description

INR

FE

Total(*)

Technical specifications

Indian rupees

Foreign exchange

A

Major Items

1

Scraper traps and pig signalling

Rated capacity as per supplier

2

Metering system

3

Cartridge filters

4

Fire protection system

5

Power supply arrangement

6

Compressors

Rated capacity as per supplier

7

Any other item (details to be given)

Subtotal

B

Bulk Materials

1

Piping

2

Electrical

3

Instrumentation

4

Any other item (details to be given)

Subtotal

C

Spares (details to be given)

Subtotal

D

Erection

1

Mechanical

2

Electrical

3

Instrumentation

4

Civil Works

5

Land

6

Any other item (details to be given)

Subtotal

E

Taxes and duties

1

Ocean freight

2

Customs duty

3

Port handling

4

Inland freight

5

Excise & CST

6

Works contract tax

7

Insurance

8

Any other applicable tax and duty (give details)

Subtotal

Total (A to E)

(*) INR + FE converted to INR

Attachment 1(b)

[see regulation 5]

Capital cost of compressors

Sr. No.

Description

INR

FE

Total(*)

Technical specifications

Indian rupees

Foreign exchange

A

Major Items

1

Compressors

Rated capacity as per supplier

2

Scraper traps and pig signalling

3

Metering system

4

Cartridge filters

5

Fire protection system

6

Power supply arrangement

7

Any other item (details to fee given)

Subtotal

B

Bulk Materials

1

Piping

2

Electrical

3

Instrumentation

4

Any other item (details to be given)

Subtotal

C

Spares (details to be given)

Subtotal

D

Erection

1

Mechanical

2

Electrical

3

Instrumentation

4

Civil Works

5

Land

6

Any other item (details to be given)

Subtotal

E

Taxes and duties

1

Ocean freight

2

Customs duty

3

Port handling

4

Inland freight

5

Excise & CST

6

Works contract tax

7

Insurance

8

Any other applicable tax and duty (give details)

Subtotal

Total (A to E)

(*) INR + FE converted to INR

Attachment 1(c)

[see regulation 5]

Capital cost of intermediate terminal

Sr. No.

Description

INR

FE

Total(*)

Technical specifications

Indian rupees

Foreign exchange

A

Major Items

1

Scrapper traps and pig signallar

Rated capacity as per supplier

2

Metering system

3

Cartridge filters

4

Fire protection system

5

Power supply arrangement

6

Any other item (details to fee given)

Subtotal

B

Bulk Materials

1

Piping

2

Electrical

3

Instrumentation

4

Any other item (details to be given)

Subtotal

C

Spares(details to be given)

Subtotal

D

Erection

1

Mechanical

2

Electricals

3

Instrumentation

4
Civil Works
5
Land
6
Any other item (details to be given)
Subtotal
E
Taxes and duties
1
Ocean freight
2
Customs duty
3
Port handling
4
Inland freight
5
Excise & CST
6
Works contract tax
7
Insurance
8
Any other applicable tax and duty (give details)
Subtotal
Total (A to E)
(*) INR + FE converted to INR
Attachment 1(d)
[see regulation 5]
Capital cost of delivery terminal
Sr. No.
Description
INR
FE
Total(*)
Technical specifications
Indian rupees
Foreign exchange
A
Major Items
1
Scraper traps and pig signalling
Rated capacity as per supplier
2
Metering system
3
Cartridge filters
4
Fire protection system
5

Power supply arrangement

6

Any other item (details to fee given)

Subtotal

B

Bulk Materials

1

Piping

2

Electrical

3

Instrumentation

4

Any other item (details to be given)

Subtotal

C

Spares(details to be given)

Subtotal

D

Erection

1

Mechanical

2

Electricals

3

Instrumentation

4

Civil Works

5

Land

6

Any other item (details to be given)

Subtotal

E

Taxes and duties

1

Ocean freight

2

Customs duty

3

Port handling

4

Inland freight

5

Excise & CST

6

Works contract tax

7

Insurance

8

Any other applicable tax and duty (give details)

Subtotal

Total (A to E)

(*) INR + FE converted to INR

Attachment 1(e)

[see regulation 5]

Capital cost of pipelines

Sr.No.

Description

INR

FE

Total(*)Technical specifications

A

Major items

1

Survey and soil investigation

2

Land ROU and crop compensation

3

Line pipes

Outer diameter and wall thickness

4

Line materials

5

Coating and pipe transportation

6

Pipeline laying

7

Cathodic protection

8

OFC, telecom and telesupervisory system

9

SCADA & APPS system

10

Power source

11

Any other item (details to be given)

Subtotal

B

Minor items

1

Ocean freight

2

Customs duty

3

Port handling

4

Inland freight

5

Excise & CST

6

Works contract tax

7

Insurance

8

Any other cost (details to be given)

Subtotal

Total (A+B)

(*) INR + FE converted to INR

Attachment 1(f)

[see regulation 5]

Profit and Loss Account (\$)

Rs.

Sr. No.

Description

Year 1

Year 2

Year 3

Year 4

.....

Year 25

1

Total inflows

a

Revenue from natural gas pipeline tariff

2

Total operating cost outflows

a

Consumables and chemicals

b

Utilities- (Power, fuel and water)

c

Salaries

d

Repair and maintenance

e

Gen administrative expenses (including bank charges on bond and performance bond as per grant of authorization) .

f

Any other expenses (to be detailed)

g

Insurance (including on line-pack volumes)

3

Profit before depreciation interest and tax (PBDIT) (1-2)

4

Interest on working capital

borrowings to the extent utilised in working capital requirement for above expenses (with rates)

5

Interest on term loans (with rates)

6

Profit before depreciation and Sax (PBDT) (3-4-5)

7

Depreciation on SLM basis and miscellaneous expenses Written off (give details of each)

8

Profit before tax (PBT) (6-7)

9

Tax provision (provide as per Income tax rules in force)

10

Profit after tax (PAT) (8-9)

Cash from operation (CFO) (10+7) - To be

transferred to cash flow sheet

(\$) Actuals or estimates, wherever applicable.

(*) Excluding all direct indirect and allocated common cost beyond transportation of natural gas in natural gas pipeline.

Financial costs and selling expenses to be excluded.

Note

1. The natural gas pipeline tariff shall be based on the actual volumes of natural gas transported in natural gas pipeline by DCF methodology adopted by considering a PIRR, at which level the net inflows equal the net outflows over the economic life of the project.

2. For expansion or extension of natural gas pipeline - data in the above format as applicable for project post expansion or extension.

3. Non-operating income (such as, interest on interest-free refundable security deposit from PNG domestic customers) not to be considered as project inflows.

4. Modvat benefit, if available, is to be reflected separately alongwith the calculations.

5. Copy of Audited Profit and Loss Account to be attached, wherever applicable.

Attachment 1(g)

[see regulation 5]

Cash flow statement

Rs.

Construction period

Economic Life

Sr.No.

Year

Year-3

Year-2

Year-1

Year 1

Year 2

Year 3

Year 4

Year 25

1

Sources of funds

a

Equity or Preference Equity (*)

b

Term Loans and short term loans

c

Cash from operations

d

Bank borrowings for working capital

e

Others (please specify)

2

Uses of funds

a

Capital expenditure

b

Normal / additional capital expenditure

c

Increase or (decrease) in net working capital

d

Repayment of loans

e

Dividends to equity holders

f

Tax on distributable profits

g

Others (please specify)

3

Cash surplus or (deficit)

a

Opening cash balance

b

Closing cash balance

(*) Redemption details to be given.

Attachment 1(h)

[see regulation 5]

Balance Sheet (\$)

Rs.

Construction period

Economic Life

Year

Year-3

Year-2

Year-1

Year 1

Year 2

Year 3

Year 4

Year 25

Assets

Gross fixed assets

Less:Accumulated depreciation (*)

Net fixed assets

Net current assets

Cash and bank balance

Investments

Miscellaneous fixed assets

Deferred revenue expenditure

Others (please specify)

Liabilities

Net worth

Equity capital

Preference share capital (**)

Reserves

Debt

Rupee term loans

FC term loans

Bank borrowings for working capital

Others (please specify)

(\$) Actuals or estimates, wherever applicable

(*) Asset head-wise depreciation calculations to be provided.

(**) To the extent non-redeemable, redeemable preference capital to be shown under debt.

Note - (a) Copy of audited Balance Sheet to be attached, wherever applicable.

Attachment 1(i)

[see regulation 5 and clause 8 and clause 9 of Schedule A]

Part I ? Computations for natural gas pipeline tariff based on project IRR.

Construction period

Economic Life

Year

Year (-) 3

Year (-) 2

Year (-) 1

Year 1

Year 2

Year 3

Year 4

Year 5

Year 6

Year 7

Year 8

Year 9

Year 10

Year 25

(a) Capital and operating cost outflows (including margin money)

(b) Inflows (Profit before depreciation, interest and income tax as per item 3 of estimated profit and loss account in Attachment 1(f))

Net Inflow / (outflows)

Cumulative inflows / (outflows)

Levelized unit natural gas pipeline tariff calculated based on the reasonable rate of return as specified in clauses 2 and 3 of Schedule A for :

a) the period of initial unit natural gas pipeline tariff (T1)

T1 : For Year 1

b) the period of five consecutive years after the initial unit natural gas pipeline tariff (T2)

T2 for year 2 to year 6

c) the period of each of the tariff review period (T3 and onwards) calculated based on reasonable rate of return as

project's internal rate of return as specified in clause 2 of Schedule A.

T3 for year 7 to year 11

? And so on ...

Notes :

1. The calculations for natural gas pipeline tariff alongwith the date are required to be submitted by the entity to the Board each time the same is due for fixation or review.
2. The adjustments required in the DCF calculations for the variations in capital cost operating cost and volumes as per the basis specified under 9 of schedule A shall be clearly specified at the time of submission of such computations to the Board.
3. Capital outflows includes outflows on account of capital work-in-progress.

Part II ? Apportionment of unit natural gas pipeline tariff over natural gas pipeline tariff zones.

[see regulation 5 and clause 8 and clause 9 of Schedule A]

(in Rs. / MMBTU)

Period applicable for initial unit natural gas

pipeline tariff / Review period

Unit natural gas pipeline tariff

Tariff zone 1

Tariff zone 2

Tariff zone3

Tariff zone4

Tariff zone5

?.till last tariff zone...

X% = [indicate one number towards %age increase over Tariff zone 1]

Y% = [indicate %age increase (lower than 100%) over preceding tariff zone]

TZ1 = [one number to be indicated for the period applicable to initial unit natural gas pipeline tariff]

Incremental increase over TZ1 = $[TZ1 \times \% / 100]$

Incremental increase over TZ1 = $Y\% \times X\% \times TZ1$

Incremental increase over TZ1 = $Y\% \times Y\% \times X\% \times TZ1$

Incremental increase over T1 = $Y\% \times Y\% \times Y\% \times X\%$

X TZ1

...and so on...

Unit natural gas pipeline tariff as per part I

For the period applicable to initial unit natural gas pipeline tariff

$TZ2 = TZ1 + \text{Above incremental increase over TZ1}$

$TZ3 = TZ2 + \text{Above incremental Increase over TZ1}$

$TZ4 = TZ3 + \text{Above incremental increase over TZ1}$

$TZ5 = TZ4 + \text{Above}$

incremental increase over

TZ1

...and so on...

Unit natural gas pipeline tariff as per Part I

Second (*)

Unit natural gas pipeline tariff as per Part I

Third (*)

:

:

Unit natural gas pipeline tariff as per Part I

Last (*)

(*) The process shall be followed on the same basis as described for the period application for initial natural gas pipeline tariff period.

(in Rs. / MMBTU)

Period

Tariff zone 1

Tariff zone 2

Tariff zone 3

Tariff zone 4

last Tariff zone

Total

Volumes considered in each tariff zone

Initial Period

Volumes considered in each tariff zone

First Review Period

Volumes considered in each tariff zone

Second Review Period

:

:

Volumes considered in each tariff zone

Last Review Period

Note :

1. The calculations for apportionment of natural gas pipeline tariff over natural gas pipeline tariff zones are required to be submitted for every tariff review period.
2. The basis followed in bifurcating the unit natural gas pipeline tariff over each of the natural gas pipeline tariff zone to be specified.
3. The basis for considering the volumes in each natural gas pipeline tariff zone to be provided (including copies of contracts to be enclosed).
4. Volumes in MMSCM to be provided in respect of above and the basis for conversion of volume of natural gas into energy equivalence in MMBTU to be indicated.

Attachment 1(j)

[see regulation 5]

Assumptions

1. Exchange rate used for determining the capital costs in Indian currency to be indicated.
2. Format for providing annual operating cost by the entity.

Description of cost elements

Rs.

A.

Variable cost

1

Consumables

Fuel (quantity and rates)

2

Utilities (quantity and rates)

a)

Power (*)

b)

Water

c)

Any other expenses (to be specified)

Sub Total

B.

Fixed and semi variable cost

1

Salary and wages

2

General administration

3

Repairs and maintenance

4

Any other expenses (to be specified) (£)

Sub Total

C

Insurance (%)

Total (A+B+C)

(*) Break-up of purchased power (over fixed charge & variable charge) & own generation to be provided.

(\$) Break-up over routing & turnaround as well as function-wise over electrical civil mechanical, etc. to be provided.

(£) Basis of direct & indirect administrative expenses (alongwith basis of allocation to be provided).

(%) Break-up of premium on loss of profit policy, facilities & line-fill quantity to be provided.

Note :

- a. For each items of operating cost, detailed assumptions and workings to be attached.
- b. For common overheads, complete basis of apportionment and allocation of costs to be provided.
3. Project implementation schedule to be given along with break up of expenditure for each year.
4. Loan schedule to be provided for each type of loan, interest rates, repayments etc.
5. Depreciation schedule as per the Companies Act and Income Tax Act should be provided.
6. Section-wise capacity of the natural gas pipeline alongwith length of each section.
7. Pipeline design capacity and rated capacity of compressors to be provided.
8. Capacity utilisation proposed and the basis to be provided.
9. Implementation period of project in phases to be provided.
- 10.

Economic life of the project.

11.

Applicable income tax rates and the exemption, if any available under the Income Tax Act, 1961 to be provided.

12.

Details of volumes of natural gas estimated to be transported in the natural gas pipeline over the economic life alongwith copies of the contracts for booking pipeline capacity to be provided in following format :

(MMSCMD)

{|

|-

|

| Year 1

| Year 2

| Year 3

| Year 4

| ----

| Year 25

|-

| Own Requirement

|

|

|
|
|
|
|
|-
| Firm Contracted

|
|
|
|
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|
|
|-
| Extra capacity (*)

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|
|-
| Total

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|
|}

(*) to be made available on common carrier open access and non-discriminatory basis

13.
Basis for conversion of natural gas volumes into its energy equivalence in MMBTU to be provided.

14.
All assumptions with reference to the estimates of capital or operating costs and any other assumption which has a material impact on the determination of the unit natural gas pipeline tariff are to be separately indicated along with the financial impact of the same.

Attachment 2 to Schedule A

[see clause 4 (3)(a) of the Schedule]

Treatment of a fixed asset in the determination of return on total capital employed for natural gas pipeline tariff.

The basis of considering fixed assets in a natural gas pipeline in the determination of the return on total capital employed shall be as per the following norms :-

(1)
A fixed asset in a natural gas pipeline is a tangible asset having a useful operating life of more than one year and is integral to the generation of revenues through natural gas pipeline tariff. Investment in securities, goodwill, current assets, accumulated loss not written-off, work-in-progress, etc. are not fixed assets.

(2)
Any change in the historical cost of the fixed asset due to revaluation or capitalization of losses shall not be considered. However, cost incurred in improvements, modification, expansion or replacement of any fixed asset shall be considered in line with the treatment prescribed in the mandatory accounting of The Institute of Chartered Accountants of India.

(3)
Only the cost of land purchased and used for putting the facilities essential to the natural gas pipeline shall be

considered. Land purchased for any future use (such as for putting-up facilities required for expansion of capacity in natural gas pipeline or its extension) shall be considered only when used.

(4)

Any change in the value of the fixed asset due to capitalization of interest on loan (including foreign exchange variation in case of any loan in any foreign currency) during the pre-commissioning phase of the fixed asset shall be considered provided such loan was utilized for creation of the said fixed asset only.

(5)

A fixed asset shall be considered for return on capital employed on a "rolling basis" till the end of the economic life of the natural gas pipeline project, provided it is not de-commissioned. On de-commissioning of the fixed asset, the value realized on its sale or scrapping shall be considered as a project inflow in the DCF calculations. In the terminal year of the economic life, the residual value of the fixed asset, which is the difference between the original cost less the amount of accumulated depreciation shall be treated as a project inflow in the DCF calculations for natural gas pipeline tariff.

(6)

Treatment of line-pack volume in natural gas pipeline shall be as per the procedure indicated below :-

(a)

Assessment of the volume of natural gas required as line pack in the natural gas pipeline as and when commissioned shall be as specified in the relevant regulations for technical standards and specifications, including safety standards.

(b)

The line-pack value of natural gas in natural gas pipelines and included in the total capital employed or determination of the unit natural gas pipeline tariff shall be considered as a non-depreciating fixed assets and the value to remain fixed over the economic life of the project.

(c)

The value of the line-pack volume shall be derived by multiplying the volume of line-pack by the average cost of natural gas at the point of injection of natural gas into the natural gas pipeline at the time of commissioning of the natural gas pipeline.

(d)

The salvage value of the line-pack volume at the end of the economic life of the project shall be equal to its value assessed at the time of its capitalization specified under sub-clause (b) and shall be considered as a project inflow.

[Schedule B]

[Inserted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020 (w.e.f. 20.11.2008).]

[See regulations 5A]

Part A ? . Procedure for determination of Unified Tariff:

1. Unified tariff shall be determined by the Board in respect of the national gas grid system for each financial year before start of such financial year. Financial year for this purpose shall mean a period commencing on the first day of April and ending on thirty-first day of March immediately following it:

Provided that in the event the date on which the Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Second Amendment, Regulation, 2020 comes into force is between first April and thirtieth September, the first unified tariff shall be determined for the period commencing from the date on which the said regulations of 2020 comes into force and ending on the immediately following thirty-first March.

Provided further that in the event, the date on which the Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Second Amendment, Regulation, 2020 comes into force is between first October and thirty-first March, the first unified tariff shall be determined for the period commencing from the date on which the said regulations of 2020 comes into force and ending on the thirty-first March next from the one immediately following. That is to say that the first unified tariff could be determined for a period ranging between six months to eighteen months.

2. The Board may determine revised unified tariff during the financial year in case it deems it necessary to do so. Without prejudice to the generality of the foregoing provision, the Board may do so in case any new interconnected pipeline is added or an existing interconnected pipeline is removed, or approved tariff in respect of any interconnected pipeline is revised, or the quantity of gas estimated to be transported including ship or pay quantity considered by the Board for determining the unified tariff undergoes a significant change.

3. Unified tariff shall be the weighted average of approved zonal tariffs in respect of all the pipelines forming part of the

national gas grid system, considering the summation of quantity of gas transported and the quantity of ship or pay for which the pipeline entity is contractually entitled to recover approved tariff for each such pipeline.

Provided that the quantities of gas transported and quantities of ship or pay through multiple pipelines shall be considered only once.

4. The Board shall determine the unified tariff applying -

(i)
the zone-wise approved tariffs applicable on the date of determination; and

(ii)
the quantity of gas estimated by it to be transported including ship or pay quantity by each pipeline forming part of the national gas grid system during the financial year for which such tariff is being determined and while making such estimates, the Board may consider the actual quantities during the twelve months preceding the date of determination, for which data may have been made available by the pipeline entities, the estimates of the quantities which may have been submitted by the pipeline entities and any other factor as it may deem fit. Such estimation by the Board shall be final and binding on all, including pipeline entities and shippers. The impact of the difference, if any, between the estimates considered by the Board and the actual quantities, shall be adjusted by the Board in the next determination of the unified tariff.

5.

(1)
Each pipeline entity shall -

(i)
provide to the Board monthly zone-wise information of approved tariff, actual volume and quantity of gas transported and the quantity of ship or pay (expressed in terms of standard cubic meters (SCM) and MMBTU and the calorific value of such gas) for each such natural gas pipeline forming part of the national gas grid system. Such information shall be provided within a period of thirty days from the date of the commencement of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Second Amendment Regulations, 2020, for immediately preceding twelve calendar months and month-wise estimates of the data-points described in this clause (other than estimates for the ship or pay quantities) for a period of next twelve calendar months, from the end of that month.

(ii)
continue to provide to the Board information described in clause (i) for each calendar month within fifteen days from the end of that month;

(iii)
provide to the Board estimates of the data-points described in clause (i) (other than estimates for the ship or pay quantities) for the next twelve months from the end of each calendar month, on a rolling basis, within fifteen days from the end of that month and the said information shall also include the data on unified zone-wise volumes and quantities of gas after the commencement of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Second Amendment Regulations, 2020.

(iv)
provide every month reasons for variation (either positive or negative exceeding five per cent) between estimates and actuals of previous month; and

(v)
submit such other one-time or periodic regular information as may be required by the Board from time to time.

(2)
The Board may issue such directions to the pipeline entities as it may deem fit regarding the manner in which information described in sub-paragraph (1) shall be submitted by them, including specifying or modifying formats, designating emails for submission of the information to the Board and specifying the portals on which the information is to be uploaded by the entities.

(3)
In case of non-submission of any information by any entity within the specified time in this regard in this paragraph, the Board may consider the information as per its best judgement for the purpose of determination of unified tariff.

6. In terms of a formula, unified tariff shall be determined is provided in the following table, namely: -

UFT

=

$$\frac{\{((AT1*EEV1) + (AT2*EEV2) + (AT3*EEV3) + \dots + (ATn*EEVn)) + Adj\}}{(EEV1 + EEV2 + EEV3 + \dots + EEVn)} - EDVn\}$$

Where:

UFT

=

unified tariff

AT1.....ATn

=

approved tariff in respect of pipelines 1 to n

applicable on the date of determination

EEV1.....EEVn

=

Quantity of natural gas estimated to be transported plus estimated ship or pay quantity for pipelines 1 to n during the financial year for which unified tariff is being determined

Adj

Impact of the difference, if any, between the estimated quantities considered by the Board and the actual quantities during the previous determination of the unified tariff

EDVn

=

Quantities of natural gas estimated to be transported plus estimated ship or pay quantities through more than one pipelines during the financial year for which unified tariff is being determined

*

:

Denotes multiplication sign

/

:

Denotes division sign

Notes: - (i) the approved tariff for pipelines 1 to n, as referred above table, denotes zone-wise approved tariff for such pipelines. Similarly, the quantities of natural gas estimated to be transported including ship or pay quantity for pipeline 1 to n, as referred to above, denotes the estimated quantity of gas transported and the estimated quantity of ship or pay for such pipelines zone;

(ii)

the formula provided in the above table shall determine the levelised unified tariff. Unified tariff for the two unified tariff zones shall be determined by the Board in a manner that the tariff for first tariff zone for unified tariff shall be 40% of the tariff for second tariff zone for unified tariff;

(iii)

in these regulations the volume of gas refers to gas expressed in terms of SCM and the quantity of gas refers to gas expressed in terms of MMBTU on GCV basis.

7. The workings of levelised unified tariff and the zonal unified tariffs are shown with the help of a numerical example in Attachment 2 to this Schedule.

Part B ? . Settlement

Difference between revenue entitlement of natural gas pipeline entities and unified tariff invoiced by such entities shall be settled amongst such pipeline entities on a periodic basis for which settlement mechanism shall be developed by

pipeline entities, facilitated by a representative from the Board within a period of two months from the date of the commencement of Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Second Amendment Regulations, 2020.

1. Revenue entitlement of natural gas pipelines that is to say the revenue entitlement for a natural gas pipeline entity in respect of a pipeline which forms part of national gas grid system shall be equal to the sum of pipeline-wise transportation charges calculated based on zone-wise actual quantity of gas transported including ship or pay quantities of each such pipeline, multiplied by the respective zone-wise approved tariff for that pipeline.

2. Amount charged by invoicing entity that is to say the amount charged by the invoicing entity in respect of natural gas pipelines which are part of national gas grid system shall be equal to the sum of amounts invoiced to the shippers, for natural gas transported including ship or pay quantities in respective unified tariff zones of each such pipeline, multiplied by the respective approved unified zonal tariff for the period.

3. In terms of a formula, entity-wise revenue entitlement and revenue charged by way of unified tariff shall be worked out as provided in the table below, namely: -

i) RE

=

$(EEV1 \times AT1 + EEV2 \times AT2 + \dots + EEVn \times ATn)$

Where:

RE

=

Revenue entitlement of the entity as per the approved tariff

EEV1.....EEVn

=

Actual quantities of natural gas transported and the quantity of ship or pay during the period by pipeline 1 to n
AT1.....ATn

=

approved tariff in respect of pipelines 1 to n for the period

ii) RU

=

$\sum AVn \times UFT$

Where:

RU

=

Tariff charged by the invoicing entity for quantities of gas evacuated at unified exit point including ship or pay quantities for the unified contractual path

AV1..... AVn

=

Actual quantities transported and the quantity of ship or pay during the period based on the unified exit points falling on the unified contractual path irrespective of number of natural gas pipelines falling on contractual path

UFT

=

unified tariff determined for the respective period

*

:

Denotes multiplication sign

Notes: The approved tariff for pipeline 1 to n, as referred to in above table, denotes the approved tariff considered for each tariff zone for pipeline 1 to n. Similarly, the actual quantities of natural gas transported and the quantity of ship or pay by pipeline 1 to n, as referred to in above table, denotes the actual gas transported and the quantity of ship or pay in each of such zones.

4. The workings of revenue entitlement and tariff invoiced are shown with the help of a numerical example in Attachment 2 to this Schedule.

5. Settlement of difference in the revenue entitlement and amount charged between various entities shall be carried out by such entities.

6. The Board may issue such further instructions and guidelines as it may deem fit including, without limiting, on periodical submission of information by the entities, working of revenue entitlement and amount charged and inter-se settlement between the entities.

7. In case any dispute regarding interpretation arises among the entities relating to these regulations, such entities may approach the Board and Board's decision in this regard shall be final.

Attachment 1 to Schedule B

(see Attachment 2 to Schedule B)

Format 1

Format for submission of actual month-wise and pipeline-wise data

Name of the Pipeline:

Month/ Year:.....

Date

Zone

Approved Tariff (a)

Actual volume of gas transported (b)

Gross Calorific Value (c)

Actual quantity of gas transported (d)

Ship or pay volumes (e)

Total Actual Volumes (f)=(d) +(e)

Transportation tariff (g)=(f)*(a)

(INR/MMBTU)

(SCM)

(Kcal/SCM)

(MMBTU)

(MMBTU)

(MMBTU)

INR

Zone-1

Zone-2

Zone-3

Zone-n

Zone

Unified Zonal Tariff (a)

Actual volume of gas transported (b)

Gross Calorific Value (c)

Actual quantity of gas transported (d)

Ship or pay volumes (e)

Total Actual Volumes (f)=(d) +(e)

Transportation tariff (g)=(f)*(a)

(INR/MMBTU)

(SCM)

(Kcal/SCM)

(MMBTU)

(MMBTU)

(MMBTU)

INR

Zone-1

Zone-2

Format 2

Format for submission of Gross Volumes Received from Upstream Pipelines Based on the Contractual Path Tariff

(*Refer Note)

Name of the Pipeline:

Month/ Year:.....

Date

From Up-stream Pipeline

Up-stream Pipeline (a)

Actual volume of gas transported (SCM) (b)

Gross Calorific Value (Kcal/SCM) (c)

Actual quantity of gas transported (MMBTU) (d)

Quantity of ship or pay (MMBTU) (e)

Pipeline 1

Pipeline 2

Note: Respective entities owning Upstream and Downstream pipelines shall provide the information after reconciling natural gas volumes (in SCM) and quantities in MMBTU for respective pipelines with other pipeline entities, if any, within the specified timelines.

Format 2A

Format for submission of Gross Volumes delivered to Downstream Pipelines Based on the Contractual Path Tariff

(Refer Note)

Name of the Pipeline:

Month/ Year:.....

Date

To Down-stream Pipeline

Down-stream Pipeline (a)

Actual volume of gas transported (SCM) (b)

Gross Calorific Value (Kcal/SCM) (c)

Actual quantity of gas transported (MMBTU) (d)

Quantity of ship or pay (MMBTU) (e)

Pipeline 1

Pipeline 2

Note: Respective entities owning Upstream and Downstream pipelines shall provide the information after reconciling natural gas volumes (in SCM) and quantities in MMBTU for respective pipelines with other pipeline entities, if any, within the specified timelines.

Format 3

Actual gas volumes transported to end consumers from the Unified Entry Point to Unified Exit Point and the ship or pay volumes for the calculation of zonal unified tariff

Name of the Pipeline:

Month/ Year:.....

Date

Unified Tariff Zone*

Actual Volumes (MMSCM)

Gross Calorific Value (Kcal/SCM)

Actual Volumes (MMBTU)

Ship or pay Volumes (MMBTU)

Zone 1

Zone 2

*Refer definition of tariff zones i.e. "tariff zones" for Unified Tariff

Note: Above formats shall be used for providing estimated quantities also.

Attachment 2 to Schedule B

{See note 4 to paragraph 3 of Part B to Schedule B}

Example for Determination of Unified Tariff, Revenue Entitlement, Amount Charged and Settlement of differences within entities

Data to be submitted in format 1.

Table 1

Approved Tariff (a)

Actual volume of gas transported (b)

Gross Calorific Value (c)

Actual quantity of gas transported (d)

Ship or pay volumes (e)

Total Actual Volumes (f)=(d) +(e)

Transportation tariff (g)=(f)*(a)

Pipelines

Zones

(INR/ MMBTU)

(SCM)

(Kcal/SCM)

(MMBTU)

(MMBTU)

(MMBTU)

INR

Pipeline A

Zone-1

10.00

30,000

9,500.00

1,130.95

200.00

1,330.95

13,309.52

Zone-2

20.00

40,000

9,400.00

1,492.06

1,492.06

29,841.27

Zone-3

30.00

50,000

9,300.00

1,845.24

1,845.24

55,357.14

Zone-4

40.00

80,000

9,600.00
 3,047.62
 100.00
 3,147.62
 1,25,904.76
 Total
 2,00,000
 7,515.87
 300.00
 7,815.87
 2,24,412.70
 Pipeline B
 Zone-1
 10.00
 45,000
 9,300.00
 1,660.71
 1,660.71
 16,607.14
 Zone-2
 11.00
 42,000
 9,400.00
 1,566.67
 1,566.67
 17,233.33
 Zone-3
 12.00
 38,000
 9,500.00
 1,432.54
 150.00
 1,582.54
 18,990.48
 Total
 1,25,000
 4,659.92
 150.00
 4,809.92
 52,830.95
 Pipeline C
 Zone-1
 30.00
 58,000
 9,546.00
 2,197.10
 2,197.10
 65,912.86
 Zone-2
 40.00
 34,000

9,645.00
 1,301.31
 1,301.31
 52,052.38
 Zone-3
 45.00
 30,000
 9,852.00
 1,172.86
 350.00
 1,522.86
 68,528.57
 Zone-4
 50.00
 68,000
 9,900.00
 2,671.43
 2,671.43
 1,33,571.43
 Zone-5
 55.00
 85,000
 9,500.00
 3,204.37
 150.00
 3,354.37
 1,84,490.08
 Total
 2,75,000
 10,547.06
 500.00
 11,047.06
 5,04,555.32
 Data to be submitted in format 2
 Data for Inter-pipeline volume flow
 Table 2
 Data by Pipeline A
 From Up-stream P/L
 Up-stream P/L
 Actual Volumes
 Gross Calorific Value
 Actual Volumes
 Ship or pay Volumes
 (SCM)
 (Kcal/SCM)
 (MMBTU)
 (MMBTU)
 Pipeline C
 1,825.00
 9,500.00
 68.80

0.50
 To Down Stream P/L
 Down-stream P/L
 Actual Volumes
 Gross Calorific Value
 Actual Volumes
 Ship or pay Volumes
 (SCM)
 (Kcal/SCM)
 (MMBTU)
 (MMBTU)
 Pipeline B
 549.00
 9,500.00
 20.70
 Data by Pipeline B
 From Up-stream P/L
 Up-stream P/L
 Actual Volumes
 Gross Calorific Value
 Actual Volumes
 Ship or pay Volumes
 (SCM)
 (Kcal/SCM)
 (MMBTU)
 (MMBTU)
 Pipeline A
 549.00
 9,500.00
 20.70
 Pipeline C
 4,392.00
 9,500.00
 165.57
 To Down Stream P/L
 Down-stream P/L
 Actual Volumes
 Gross Calorific Value
 Actual Volumes
 Ship or pay Volumes
 (SCM)
 (Kcal/SCM)
 (MMBTU)
 (MMBTU)
 NIL
 Data by Pipeline C
 From Up-stream P/L
 Up-stream P/L
 Actual Volumes
 Gross Calorific Value
 Actual Volumes

Ship or pay Volumes

(SCM)

(Kcal/SCM)

(MMBTU)

(MMBTU)

Nil

To Down Stream P/L

Down-stream P/L

Actual Volumes

Gross Calorific Value

Actual Volumes

Ship or pay Volumes

(SCM)

(Kcal/SCM)

(MMBTU)

(MMBTU)

Pipeline A

1,825.00

9,500.00

68.80

0.50

Pipeline B

4,392.00

9,500.00

165.57

Summary of the duplicate volume flows

Pipeline From

Pipeline to

Actual Volumes (SCM)

Gross Calorific Value (Kcal/SCM)

Actual Volumes (MMBTU)

Ship or pay Volumes (MMBTU)

Pipeline C

Pipeline A

1,825.00

9,500.00

68.80

0.50

Pipeline A

Pipeline B

549.00

9,500.00

20.70

0.00

Pipeline C

Pipeline B

4,392.00

9,500.00

165.57

0.00

Total

6,766.00

255.07

0.50

Data to be submitted in format 3

Data for Unified Zonal volume flows to be provided by Entities:

Data provided by Pipeline A

Table 3

Unified Tariff Zone*

Actual Volumes (SCM)

Gross Calorific Value (Kcal/SCM)

Actual Volumes (MMBTU)

Ship or pay Volumes (MMBTU)

Zone 1

30,000.00

9,500.00

1,130.95

200.00

Zone 2

1,67,626.00

9,464.21

6,295.42

99.50

Data provided by Pipeline B

Unified Tariff Zone*

Actual Volumes (SCM)

Gross Calorific Value (Kcal/SCM)

Actual Volumes (MMBTU)

Ship or pay Volumes (MMBTU)

Zone 1

45,000.00

9,300.00

1,660.71

-

Zone 2

80,000.00

9,447.50

2,999.21

150.00

Data provided by Pipeline C

Unified Tariff Zone*

Actual Volumes (SCM)

Gross Calorific Value (Kcal/SCM)

Actual Volumes (MMBTU)

Ship or pay Volumes (MMBTU)

Zone 1

53,608.00

9,549.77

2,031.52

Zone 2

2,17,000.00

9,696.73

8,349.96

500.00

Consolidated Volumes

Unified Tariff Zone*

Actual Volumes (SCM)

Gross Calorific Value (Kcal/SCM)

Actual Volumes (MMBTU)

Ship or pay Volumes (MMBTU)

Zone 1

1,28,608.00

9,450.77

4,823.19

200.00

Zone 2

4,64,626.00

9,569.93

17,644.59

749.50

Table 4

Calculation of Unified Tariff

Pipelines

Revenue Entitlement

Quantity for Revenue Entitlement

Duplicate Volumes

UFT

INR

in MMBTU

in MMBTU

Rs./ MMBTU

(a)

(b)

(c)

a/(b-c)

Pipeline A

2,24,412.70

7,815.87

255.57

33.39

Pipeline B

5,04,555.32

11,047.06

Pipeline C

52,830.95

4,809.92

Total Revenue

7,81,798.97

23,672.85

Calculation of amount charged by entities way of charging unified tariff:

Table 5

Amount charged by Pipeline A

Zone

Unified Zonal Tariff

Actual volume of gas transported

Gross Calorific Value

Actual quantity of gas transported

Ship or pay volumes

Total Actual Volumes

Transportation tariff

(a)

(b)

(c)

(d)

(e)

(f)=(d) +(e)

(g)=(f)*(a)

(INR/MMBTU)

(SCM)

(Kcal/SCM)

(MMBTU)

(MMBTU)

(MMBTU)

INR

Zone 1

X

30,000.00

3,769.84

1,130.95

200.00

1,330.95

Zone 2

Y

1,67,626.00

3,755.64

6,295.42

99.50

6,394.92

Total

7,426.38

299.50

7,725.88

Amount charged by Pipeline B

Zone

Unified Zonal Tariff

Actual volume of gas transported

Gross Calorific Value

Actual quantity of gas transported

Ship or pay volumes

Total Actual Volumes

Transportation tariff

(a)

(b)

(c)

(d)
 (e)
 (f)=(d) +(e)
 (g)=(f)*(a)
 (INR/MMBTU)
 (SCM)
 (Kcal/SCM)
 (MMBTU)
 (MMBTU)
 (MMBTU)
 INR
 Zone 1
 X
 45,000.00
 3,690.48
 1,660.71
 -
 1,660.71
 Zone 2
 Y
 80,000.00
 3,749.01
 2,999.21
 150.00
 3,149.21
 Total
 4,659.92
 150.00
 4,809.92
 Amount charged by Pipeline C
 Zone
 Unified Zonal Tariff
 Actual volume of gas transported
 Gross Calorific Value
 Actual quantity of gas transported
 Ship or pay volumes
 Total Actual Volumes
 Transportation tariff
 (a)
 (b)
 (c)
 (d)
 (e)
 (f)=(d) +(e)
 (g)=(f)*(a)
 (INR/MMBTU)
 (SCM)
 (Kcal/SCM)
 (MMBTU)
 (MMBTU)
 (MMBTU)

INR

Zone 1

X

53,608.00

3,789.59

2,031.52

-

2,031.52

Zone 2

Y

2,17,000.00

3,847.91

8,349.96

500.00

8,849.96

Total

10,381.48

500.00

10,881.48

Statement of settlement of differences among entities:

Table 6

Pipelines

Revenue Entitlement

Amount Charged (from table 5)

Amount to be Paid/ (Received)

(a)

(b)

(b-a)

Pipeline A

2,24,412.70

Pipeline B

52,830.95

Pipeline C

5,04,555.32

Total

7,81,798.97

7,81,798.97

0.00

[Schedule C]

[Substituted by Notification No. F. No. PNGRB/COM/2-NGPL/Tariff (3)/2014 Vol-IV(Part-1) (P-1439), dated 23.11.2020.]

{See regulation 2(1) (ec)}

List of natural gas pipelines which are part of national gas grid system for calculation of Unified Tariff

S.No.

Entity Authorized

Pipeline Name

1

GAIL (India) Limited

Hazira-Vijaipur-Jagdishpur-GREP-Dahej-Vijaipur,

Dahej-Uran-Dabhol-Panvel

Natural Gas Pipeline,

Dahej-Vijaipur

(DVPL)-Vijaipur-Dadri (GREP) capacity augmentation,
Jagdishpur-Haldia-Bokaro-Dhamra
Natural Gas Pipeline network,
Dadri-Bawana-Nangal
Natural Gas Pipeline,
Chainsa-Jhajjar-Hissar
Natural Gas Pipeline,
Dabhol-Bangalore Natural Gas Pipeline.

2

Indian Oil Corporation Limited
Dadri-Panipat.

3

Pipeline Infrastructure Limited
East West Natural Gas Pipeline.

4

Gujarat State Petronet Limited
GSPL's High Pressure Gujarat Gas Grid.

5

Reliance Gas Pipelines Limited
Shahdol-Phulpur.

6

GSPL India Gasnet Limited
Mehsana ? Bhatinda, Bhatinda - Jammu ?
Srinagar.

7

GSPL India Transco Limited
Mallavaram-Bhopal-Bhilwara-Vijaipur.
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